

AYUSH RAJ TIWARY

AI/ML Engineer | Full-Stack Developer | LLM Applications
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PROFESSIONAL SUMMARY

ML Engineering Intern at FlyRank.ai (Jul–Sep 2026) and full-stack developer with 2 deployed production AI projects. Built NEURO-CORE (EfficientNet-B0, 95.75%) and Spice Garden (8 LangChain.js agents, Cohere RAG, Razorpay, Socket.io). Experienced with PyTorch, FastAPI, Next.js, pgvector, Redis, and Docker. Google Gen AI Exchange 2025 developer.

INTERNSHIP EXPERIENCE

Machine Learning Engineering Intern — FlyRank.ai

Jul 2026 – Sep 2026

FlyRank Corp. · Remote | Tracks: Machine Learning, AI Fluency | Cert ID: FR-D11-4A8A8-87BC0

- Completed 36 practical ML assignments across data wrangling, embeddings and clustering, intent and opportunity modeling, and insight-to-action pipelines; logged 190+ verified hours including capstone project reviewed and accepted by lead mentor.
- Completed 20 verified Anthropic Academy courses and qualified independently in 2 tracks (Machine Learning, AI Fluency); capstone: Send the Link — Launch, Demo and Story.

TECHNICAL SKILLS

AI and ML: PyTorch, CNN, ResNet-18, EfficientNet-B0, Grad-CAM, Focal Loss, Transfer Learning, TensorFlow, OpenCV, MediaPipe, NumPy, Pandas, Scikit-learn

LLM and GenAI: LangChain.js, Groq API, Cohere Embeddings, pgvector, RAG Pipelines, Prompt Engineering, Multilingual NLP

Frontend: React.js, Next.js, Three.js, Socket.io, Zustand, Tailwind CSS, REST APIs, PWA

Backend: Node.js, Express.js, FastAPI, PostgreSQL, Prisma ORM, MongoDB, Redis, WebSocket

DevOps and Cloud: Docker, Supabase, Vercel, Render, Netlify, Cloudinary, Upstash, Git, GitHub, Postman

Languages: Python, JavaScript, Java, C, C++

Core CS: Data Structures and Algorithms, OOP, System Design, Model Deployment

PROJECTS

NEURO-CORE — Brain Tumor MRI Detection with Gesture Control

[GitHub](#) | [Live Demo](#)

Python · PyTorch · CNN · ResNet-18 · EfficientNet-B0 · Grad-CAM · Focal Loss · FastAPI · MediaPipe · Three.js · Groq · Render

- Trained 3 independent PyTorch models on 7,200 brain MRI scans using Focal Loss: Custom CNN (89.94%), ResNet-18 (95.12%), and EfficientNet-B0 (95.75% best — only 5.3M params vs 11.3M ResNet-18).
- Implemented per-model Grad-CAM heatmaps and integrated Groq API for LLM-generated plain-language diagnostic summaries; built in-browser MediaPipe gesture engine (WebAssembly) for fully sterile contactless review.

Spice Garden — AI-Driven Smart Dining Assistant

[GitHub](#) | [Live Demo](#)

Next.js · Node.js · PostgreSQL · pgvector · Cohere · LangChain.js · Groq · Redis · Socket.io · Prisma · Razorpay · Cloudinary · Docker · Vercel · Render

- Architected 8-agent LangChain.js system (Orchestrator, Recommendation, Upsell, Sentiment, Memory, Multilingual NLU, Group Coordinator, Order Validation) routing intents to specialized agents.
- Built Cohere RAG pipeline over pgvector with multilingual NLU (English, Hinglish, Telugu), dietary filtering, allergy alerts, and cross-session memory; 25+ features including Razorpay, WhatsApp notifications, Socket.io group ordering, and PWA; 25-model Prisma schema.

ACHIEVEMENTS

Google Gen AI Exchange 2025 — Mental Health Assistant

AI Developer

- Designed and deployed a conversational AI mental health assistant using Gen AI APIs for natural language understanding and response generation within hackathon constraints.

EDUCATION

B.Tech in Computer Science Engineering

2023 – 2027

Jharkhand Rai University, Ranchi | Specialization: AI and ML | CGPA: 7.3 (up to 6th sem) | Current: 7th sem